WORK INSTRUCTION MANUAL

Test Equipment Control System

Web-Based Automation Interface

For:

• Keithley Power Supply (Model 2230-30-1)
• Keithley Digital Multimeter (Model 6500/7510)
• Keysight Oscilloscope (Model DSOX6004A)

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1. SYSTEM OVERVIEW

Purpose: This system provides web-based control interfaces for laboratory test equipment. It allows technicians to control instruments from any computer on the same network without installing special software.

Key Features:

- • Browser-based interface - no special software required
- • Multiple users can access from different computers
- • Real-time measurements and control
- • Automatic data logging and export
- • Emergency stop features for safety

Available Control Systems:

System	Port Number	Equipment
Power Supply Control	7861	Keithley 2230-30-1
DMM Control	7862	Keithley DMM6500/7510
Oscilloscope Control	7864	Keysight DSOX6004A
Unified	7860	All

2. PREREQUISITES AND REQUIREMENTS

Network Requirements:

- • Your computer must be connected to the same network as the control PC
- • Control PC IP Address: **192.168.128.175**
- • Network subnet: 192.168.128.xxx

Browser Requirements:

Recommended browsers (use latest versions):

- • Google Chrome (Recommended)
- • Microsoft Edge
- • Mozilla Firefox
- • Safari (Mac)

Equipment Setup:

- • All instruments must be connected via USB to the control PC
- • Instruments must be powered ON before accessing the interface
- • USB drivers must be installed (already configured on control PC)
- • VISA drivers must be running (automatic on control PC)

3. SYSTEM ACCESS INSTRUCTIONS

IMPORTANT: The control PC (192.168.128.175) must be running the appropriate control program before you can access it from your computer.

Step-by-Step Access Instructions:

Step 1: Check Network Connection

- ○ • Connect your computer to the same network as the control PC
- ○ • Verify your IP is in range 192.168.128.xxx
- ○ • Test connection: Open Command Prompt and type: ping 192.168.128.175

Step 2: Access the Web Interface

Open your web browser and enter the appropriate URL:

Equipment	URL to Enter in Browser
Power Supply	**http://192.168.128.175:7861**
Digital Multimeter	**http://192.168.128.175:7862**
Oscilloscope	**http://192.168.128.175:7864**
All	**http://192.168.128.175:7860**

Step 3: Wait for Interface to Load

- ○ • The web interface should load within 5-10 seconds
- ○ • If it doesn't load, check troubleshooting section

4. KEITHLEY POWER SUPPLY CONTROL

Access URL: http://192.168.128.175:7861

4.1 Connection Setup

Step 1: Verify the VISA address is correct (pre-filled):

```
Default: USB0::0x05E6::0x2230::805224014806770001::INSTR
```

Step 2: Click the 'Connect' button
Step 3: Wait for 'Connected' status to appear

4.2 Channel Configuration

The power supply has 3 independent channels:

- • Channel 1: 0-30V, 0-3A
- • Channel 2: 0-30V, 0-3A
- • Channel 3: 0-5V, 0-3A

To Configure a Channel:

1. Go to 'Channel Controls' section
2. Select the channel tab (Channel 1, 2, or 3)
3. Set the Voltage using the slider
4. Set the Current Limit (important for safety)
5. Set the OVP (Over-Voltage Protection) level
6. Click 'Configure Ch#' button
7. Click 'Enable Output' to turn on the channel

4.3 Measurements

Single Channel Measurement:

- • Click 'Measure' button for the specific channel
- • View Voltage, Current, and Power readings

All Channels Measurement:

- • Go to 'Global Operations' section
- • Click 'Measure All Channels'
- • All three channels measured sequentially

4.4 Data Logging

1. Enable Auto-Measurement checkbox
2. Set measurement interval (seconds)
3. Click 'Export to CSV' to save data
4. File saves with timestamp in filename

4.5 Emergency Stop

EMERGENCY STOP BUTTON:

- • Located at top of interface
- • RED button labeled 'EMERGENCY STOP'
- • Immediately disables ALL outputs
- • Use if you see smoke, sparks, or any danger

5. KEITHLEY DMM CONTROL

Access URL: http://192.168.128.175:7862

5.1 Connection Setup

1. Go to 'Connection' tab
2. Verify VISA address (pre-filled)

3. Click 'Connect' button
4. Check connection status shows instrument info

5.2 Measurement Types

Measurement	Range	Use Case
DC Voltage	0.001-1000V	Battery, power supply testing
AC Voltage	0.001-750V	Mains, signal testing
DC Current	0.001µA-3A	Circuit current draw
Resistance	0.001Ω-100MΩ	Component testing

5.3 Making Measurements

Single Measurement:

1. Go to 'Measurements' tab
2. Select Measurement Function (DC_VOLTAGE, etc.)
3. Set Range (or use Auto range)
4. Set Resolution (higher = more accurate but slower)
5. Set NPLC (1.0 is standard)
6. Click 'Single Measurement'
7. Read result in display

Continuous Measurements:

1. Configure measurement settings
2. Set interval (seconds between measurements)
3. Click 'Start Continuous'
4. Click 'Stop Continuous' when done

5.4 Statistics and Analysis

- • Go to 'Statistics & Analysis' tab
- • Set number of points to analyze
- • Click 'Calculate Statistics'
- • View Mean, Std Dev, Min, Max values
- • Click 'Update Plot' for trend graph

5.5 Data Export

1. Go to 'Data Export' tab
2. Select format (CSV, JSON, or Excel)
3. Click 'Export Data'
4. File saves with timestamp

6. KEYSIGHT OSCILLOSCOPE CONTROL

Access URL: http://192.168.128.175:7864

6.1 Connection Setup

1. Go to 'Connection' tab
2. Verify VISA address
3. Click 'Connect'
4. Click 'Test' to verify connection

6.2 Channel Configuration

The oscilloscope has 4 input channels:

To Configure Channels:

1. Go to 'Channel Configuration' tab
2. Check boxes for channels to enable
3. Set V/div (vertical scale)
4. Set Offset if needed
5. Select Coupling (AC or DC)
6. Select Probe (1x, 10x, or 100x)
7. Click 'Configure Channels'
8. Click 'Autoscale' for automatic setup

6.3 Timebase and Trigger

Timebase (Horizontal) Settings:

1. Go to 'Timebase & Trigger' tab
2. Select Time/div from dropdown
3. Click 'Apply Timebase'

Trigger Settings:

1. Select trigger Source (CH1-CH4)
2. Set trigger Level (voltage)
3. Select Slope (Rising/Falling/Either)
4. Click 'Apply Trigger'

6.4 Measurements

Available measurements:

- • Frequency and Period
- • Peak-to-Peak, Amplitude
- • Average, RMS values
- • Rise/Fall times
- • Duty cycle

To Make Measurements:

1. Go to 'Measurements' tab
2. Select Source (CH1-4 or MATH1-4)
3. Select Measurement Type
4. Click 'Measure' or 'Show All'

6.5 Data Acquisition

1. Go to 'Operations & File Management' tab
2. Select channels to capture (Ch1-4)
3. Click 'Capture Screenshot' for display image
4. Click 'Acquire Data' to capture waveforms
5. Click 'Export CSV' for data file
6. Click 'Generate Plot' for graphs
7. Or click 'Full Automation' for all steps

7. TROUBLESHOOTING

Common Issues and Solutions:

Problem	Solution
Cannot access web interface	1. Check network connection 2. Ping 192.168.128.175

	3. Verify control program is running 4. Check firewall settings
Connection failed to instrument	1. Check USB cable connection 2. Verify instrument is powered ON 3. Check VISA address is correct 4. Restart the control program
Measurements show 'Error'	1. Check probe connections 2. Verify measurement range 3. Check channel is enabled 4. Try reconnecting to instrument
Interface is slow or freezing	1. Refresh browser page 2. Clear browser cache 3. Close other browser tabs 4. Check network speed
Data export not working	1. Check disk space available 2. Verify write permissions 3. Try different export format 4. Check data is acquired first

Network Diagnostics:

To test network connection:

1. Open Command Prompt (Windows) or Terminal (Mac/Linux)
2. Type: ping 192.168.128.175
3. Should see replies with time < 10ms
4. If 'Request timed out', check network cable/WiFi

8. SAFETY GUIDELINES

⚠ IMPORTANT SAFETY RULES

Electrical Safety:

- • Always set current limits before enabling outputs
- • Never exceed device voltage ratings
- • Check connections before applying power
- • Use proper probe ratings (1x, 10x, 100x)
- • Keep liquids away from equipment

Emergency Procedures:

If you observe:

- • Smoke or burning smell → Press EMERGENCY STOP
- • Sparks or arcing → Press EMERGENCY STOP
- • Unexpected behavior → Disable all outputs
- • Equipment damage → Disconnect immediately

Best Practices:

- • Start with low voltages and increase gradually
- • Always disable outputs before changing connections
- • Document your settings before experiments
- • Export data regularly to prevent loss
- • Check calibration dates on equipment

9. APPENDIX

Quick Reference Card

Function	URL	Port
Power Supply	http://192.168.128.175:7861	7861
DMM	http://192.168.128.175:7862	7862
Oscilloscope	http://192.168.128.175:7864	7864
All	http://192.168.128.175:7860	7860

Technical Support

For technical assistance:

- • Contact: Anirudh Iyengar
- • Organization: Digantara Research and Technologies
- • Control PC IP: 192.168.128.175

File Locations

Default save locations on control PC:

- • Data files: /home/____e/data/
- • Graphs: /home/____e/graphs/
- • Screenshots: /home/____e/screenshots/
- • CSV exports: Timestamped filenames

Keyboard Shortcuts

Browser shortcuts for efficiency:

- • F5: Refresh page
- • Ctrl + R: Reload interface
- • Ctrl + Shift + R: Hard refresh (clear cache)
- • F11: Full screen mode
- • Ctrl + + : Zoom in
- • Ctrl + - : Zoom out
- • Ctrl + 0 : Reset zoom

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